Welcome to VKEMS VSG - March 24th-26th

Part of Stream's Open Data Day 2026

Open Data Day (ODD) is an annual celebration of open data all over the world. Groups from any country create local events where they will use open data in their communities. ODD is maintained and supported by the Open Knowledge Foundation (OKFN) through the Open Knowledge Network.

Stream - a water industry group working to stimulate innovation and collaboration through open and shared data related to the water sector - has taken part in Open Data Day for the past few years. The event brings together brilliant minds from academia and industry to tackle some of the most pressing, system-level challenges in the water sector. By opening up data and fostering creative problem-solving, we aim to inspire fresh ideas for products and services that deliver real impact for stakeholders and the environment.

1.0 Stream headline challenge statement

This year's theme is water efficiency and you're here to work on new solutions that help address this challenge. Our headline challenge statement is:

How might we, using a variety of data sources, better understand and communicate national, regional and local water availability, so that everyone can make better-informed water use decisions?*

**individuals, households, organisations, government and regulators*

However, we understand that's quite broad so how might this translate at a more granular level?

2.0 VKEMS VSG problem statements

Problem statement 1 – Forecasting leakage

The leakage team at NWL create a forecast throughout the year on how much leakage they expect to happen week to week. This is used to support decision making for workforce planning and deployment (faster mobilisation to fix leaks) and ultimately be more efficient (less wasted resource in terms of water, people and money). These forecasts are created by averaging historical (last 5 years) leakage changes on a weekly basis.

Whilst this approach is nice and simple to run, it is not particularly accurate and there is often a gap between forecast and actual. Averaging doesn't deal very well with situations that are known as leakage breakouts (where lots of leaks happen all at once across an area – usually following a protracted cold or dry spell which causes the ground to move and fracture pipes). A breakout will create a spike in leakage but

the level of leakage generally comes down quicker than the model predicts leading to inaccurate forecasts.

The challenge is to find ways to refine the forecast to narrow down the margin of error by looking at how other factors have played a part, such as preceding (say, 3 months) weather (particularly protracted dry and cold spells). At present the forecast only relies upon historic levels, with no understanding of the context at the time. Up to 20 years of daily leakage levels are available for each company (Essex and Suffolk Water and Northumbrian Water – so 1 data point per day per supply zone) and leak repair data is also available going back about 5 years for both areas.

Problem statement 2 – Characterising district meter areas so that different strategic approaches to managing leakage can be applied appropriately

A district metered area is a discrete area of a water network that has a known boundary and flow meters that monitor the volume of water going into that area every day. On average they tend to feed between 1,000 and 2,000 properties and cover about 11km of distribution network, although there are some outliers on either side of these ranges. The DMA will have certain physical (and non-physical) characteristics that may individually, or in concert, determine whether that DMA is more or less likely to experience leakage under different sets of circumstances. Factors may include pipe material, age, ground conditions, property type, pressure (along with many more). How might we use available data to determine the character of a DMA so that different leakage management strategies can be applied appropriately? Being able to do this enables focus and intervention to be more fine-tuned, leading to better leakage performance and ultimately positive outcomes for customers, society, the environment and the water companies themselves.

3.0 What resources are available to you?

All resources will be available on the Stream community page which you will have access to during the event. You'll also see the resources we made available to participants at the Wessex Water data challenge earlier in the month.

We've identified the following data sets and supporting content for our challenge statements as detailed below. You are of course able to include other supporting content for your idea.

Problem statement 1 (forecasting leakage to support resource prioritisation)

Daily leakage number for each company (NW and E&SW) to water resource zone level going back as far as possible (possibly 20 years). Zone will be expressed as a geographical centrepoint, not a shapefile and the zone name will not be included. Note: the daily leakage number is the raw estimate and not the fully calculated and audited number.

Problem statement 2 (characterising which DMA's are likely to be troublesome for strategic prioritisation).

For a cohort of about 20 anonymised DMAs covering a mix of rural, urban, rural/ urban mix etc:

1. 15 min pressure and flow values for the last 3 years
2. No. of properties split HH / NHH
3. County name
4. Daily (raw) leakage value for the last 3 years
5. Pipe characteristics (length and material)
6. Min and Max elevation for the DMA

4.0 Links

All data and supporting information is available on the Stream platform: link will be added close of day on Monday 23rd March.

5.0 Developing your idea

Think about the overall challenge statement around water efficiency and the specific focus area your solution will address. Clearly capture how you plan to do this to keep you focused over the next few days being sure to consider the following things.

- Who is the target audience for your idea? Individuals, households, organisations, government or regulators?
- Who will benefit from your team's idea? This might extend beyond the users. Keep them front and centre in your mind to make sure your solution addresses all of their needs.
- How easy is your idea to implement - think about any technical constraints, time limitations, costs, changes to existing infrastructure etc?

6.0 What are we looking for?

We're looking for innovative ideas that address the overall challenge statement. You'll be able to demonstrate how you've used the available data sets (and others you may have access to) to create a solution for your chosen problem area.

7.0 Programme schedule

Day 1: Tuesday 24 March 2026

	Main room sessions chaired by Alan Champneys, University of Bristol
09:30-09:45	Zoom opens
09:45-10:00	Welcome
10:00-10:20	**Introduction to Challenge 1: Forecasting leakage** *Andrew Blenkharn, Northumbrian Water*
10:20-10:30	Questions and discussion
10:30-10:50	**Introduction to Challenge 2: Characterising DMAs** *Andrew Blenkharn, Northumbrian Water*
10:50-11:00	Questions and discussion
11:00-11:15	**Break**
11:15-12:30	Introduction to Mural, ideas generation and group formation
12:30-13:30	**Lunch**
13:30-16:15	Group work in breakout rooms
16:15-16:30	End of day catch-up

Day 2: Wednesday 25 March 2026

	Main room sessions chaired by Alan Champneys, University of Bristol
09:25-09:30	Zoom opens
09:30-10:00	**Pitch your ideas to the experts** *Andrew Blenkharn, Northumbrian Water* *Chad Anderson-Grout, Southern Water* *Kaustuv Nag, Southern Water*
10:00-12:10	Group work in breakout rooms

12:10-12:30	**Wessex Water Data Challenge: The Winning Idea** *Ramiz Beig Zali, ForesightAI Limited*
12:30-13:30	**Lunch**
13:30-16:15	Group work in breakout rooms
16:15-16:30	End of day catch-up

Day 3: Thursday 26 March 2026

	Main room sessions chaired by Alan Champneys, University of Bristol
09:25-09:30	Zoom opens
09:30-09:45	Start of day catch-up
09:45-12:30	Group work in breakout rooms
12:30-13:30	**Lunch**
13:30-15:00	Group work in breakout rooms
15:00-16:30	Final presentations and next steps

8.0 Want to tell people you're taking part?

We'll be using a couple of hashtags to promote the event. If you spread the word on your social channels we'd love it if you'd include these.

#OpenDataDay26 #StreamOpenDataDay26

9.0 About Stream and VKEMS

About Stream

Stream is working to stimulate innovation and collaboration through open and shared data.

Our vision is to unlock the potential of water data to benefit customers, society, and the environment and to use data to address key water sector challenges.

Our Open Data platform is designed to offer clear and accessible access to water-related data for a range of uses such as research, decision-making and raising public awareness.

This event brings together brilliant minds from academia and industry to tackle some of the most pressing, system-level challenges in the water sector. By opening up data and fostering creative problem-solving, we aim to inspire fresh ideas for products and services that deliver real impact for stakeholders and the environment.

https://www.linkedin.com/company/stream-unlocking-water-data

About VKEMS

The Virtual Forum for Knowledge Exchange (V-KEMS) is an ongoing collaboration between The International Centre for Mathematical Sciences, Isaac Newton Institute, Newton Gateway to Mathematics and Innovate UK Business Connect, and various representatives from the mathematical sciences community.

The main aim of the forum is to identify a range of virtual approaches that will help address challenges from business and industry, the third sector, and other groups outside academia. These challenges may be long-standing or may have arisen directly as a consequence of the present disruption to UK society (such as the pandemic).